

LLM Agents for Smart City Management: Enhancing Decision Support Through Multi-Agent AI Systems

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1. RESEARCH OBJECTIVE & SYSTEM ARCHITECTURE

Multi-agent architecture routing urban queries to domain-specific agent panels; evaluates response quality using G-Eval (LLM-as-a-judge).

2. BENCHMARK DATASETS & EMPIRICAL EVALUATION

Smart-city municipal management query and routing benchmark.

Key Numerical Findings & Empirical Proofs:

94%–99% routing accuracy; G-Eval decision quality scores of 0.68–0.74 vs. 0.30–0.38 for standalone LLMs (over 2x quality improvement).

3. CRITICAL ARCHITECTURAL FLAWS & METHODOLOGICAL LIMITATIONS

Identified System Failure Modes:

Uses G-Eval purely as an offline evaluation metric rather than an online runtime decision controller; centralized coordinator is a bottleneck.

4. RESEARCH GAP & RELEVANCE TO CAG-DELPHI

This work exemplifies the *Unconditional Consultation Dilemma* observed across the 15-paper corpus. By failing to incorporate epistemic uncertainty self-gating prior to multi-agent activation, the architecture incurs severe token inflation and exposes confident primary solutions to debate degeneration noise. In CAG-Delphi, this paper serves as an empirical anchor for diversity-governed consensus and comparative baseline evaluation.

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